A2: Design with basic Stresses

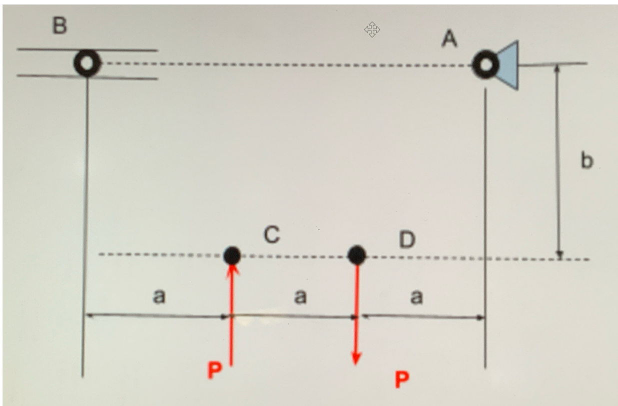
Given information:

- material = A500 Structural Steel
- Load: $P = 25\ kN$
- $a = 0.4m$ (Pin support)
- $b = 0.3m$ (Roller support)
- All truss members must have the same cross-sectinal area and shape
- All pins are the same size
- Pin material = hardened tool Steel
- Pin yield Shear Strength = 170 ksi
- Pin density = $0.278\ lb/in^3$
- Truss Safety Factor = 3.5
- Pin Safety factor = 4

Why I Chose my load Force

I chose the load force $P = 25\ kN$ because it falls in the middle of the given numbers 20 - 30 KN.

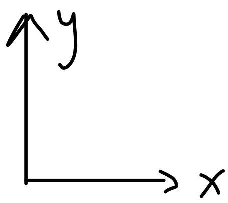
FBD/Truss Geometry

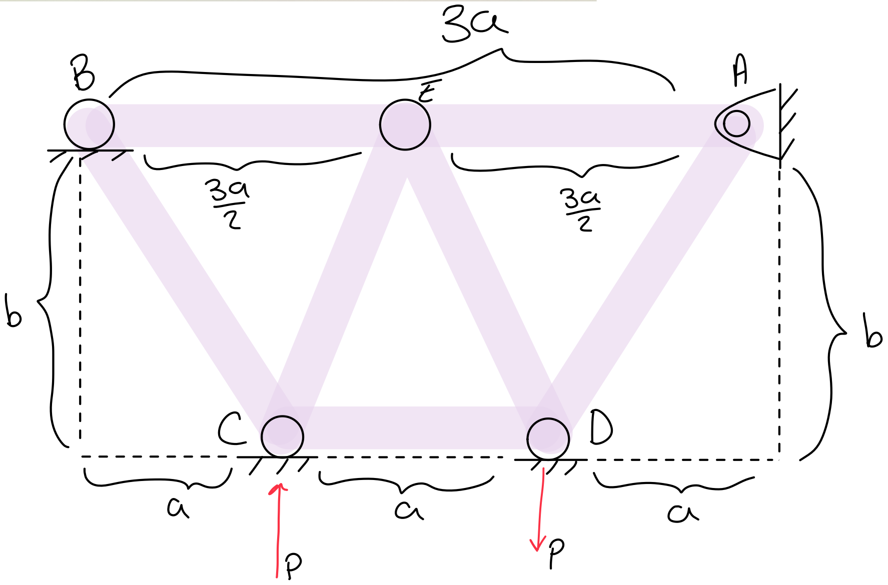


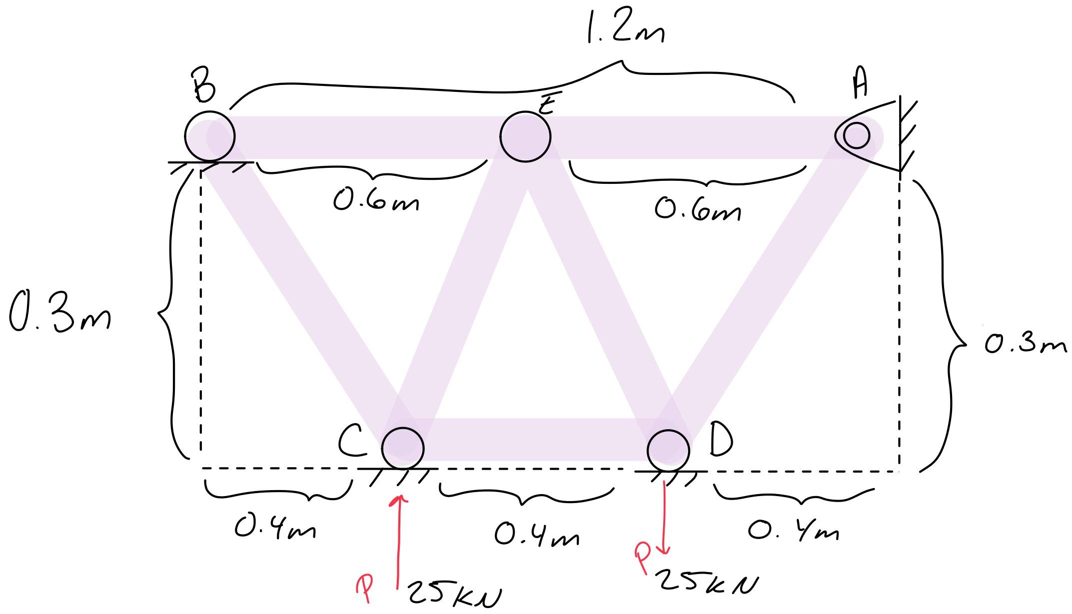
$a = 0.4m$

$b = 0.3m$

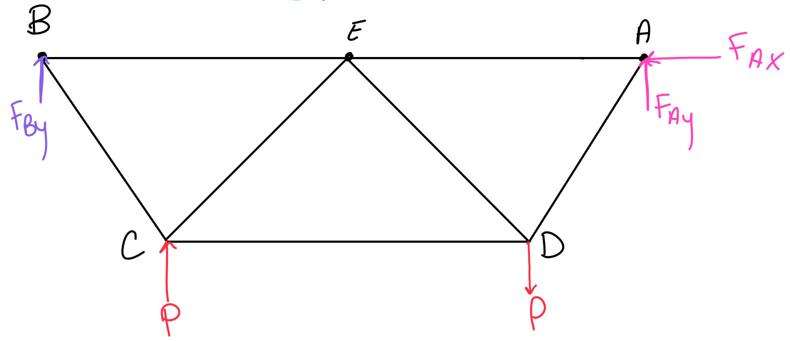
$P = 25\ KN$







FBD



Reaction Forces

$\Sigma F_x = 0 \Rightarrow \boxed{A_x = 0}$

$\Sigma M_A = 0 \Rightarrow P(0.8m) + P(0.4m) - B_y(1.2m) = 0$

$\Rightarrow 25_{kN}(0.8m) + 25_{kN}(0.4m) - B_y(1.2m) = 0 \Rightarrow 25kN(0.8m) + 25_{kN}(0.4m) = B_y(1.2m)$

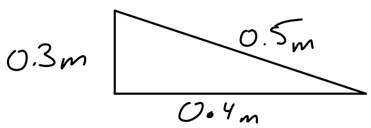
$\Rightarrow 20\,kN/m + 10\,kN/m = B_y(1.2m) \Rightarrow \frac{30\,kN/\cancel{m}}{1.2\cancel{m}} = \frac{B_y(1.2m)}{1.2m} \Rightarrow \boxed{B_y = 25\,kN}$

$\Sigma F_y = 0 \Rightarrow B_y + A_y - P - P = 0 \Rightarrow 25_{kN} + A_y \underset{+}{-} 25 \underset{+}{-} 25 = 0$

$\Rightarrow 25_{kN} + A_y = 25_{kN} + 25_{kN} \Rightarrow \begin{array}{r} 25_{kN} + A_y \\ -25_{kN} \end{array} = \begin{array}{r} 50_{kN} \\ -25_{kN} \end{array} \Rightarrow \boxed{A_y = 25\,kN}$

angles

SOH CAH TOA



$a^2 + b^2 = c^2$

$0.4^2 + 0.3^2 = c^2 \Rightarrow \sqrt{0.25} = \sqrt{c^2}$

$c = 0.5$

$\sin\theta = \frac{0.3}{0.5} = \boxed{0.6}$

$\cos\theta = \frac{0.4}{0.5} \boxed{= 0.8}$ $\quad \tan\theta = \frac{0.3}{0.4} \boxed{= 0.75}$